Traffic Pattern and Time-Space Diagram

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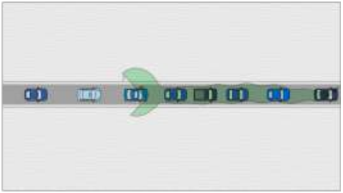
Phantom Traffic Jam

- Traffic is congested without clear causes, i.e., we cannot see clear bottlenecks, such as ramps, accidents.
- Small disturbances (a driver hitting the brake too hard, or getting too close to another car) in heavy traffic can become amplified into a self-sustaining traffic jam.
- In mathematics, it is related to traffic instability.
- Resulting in stop-and-go waves (introduced in the following section).

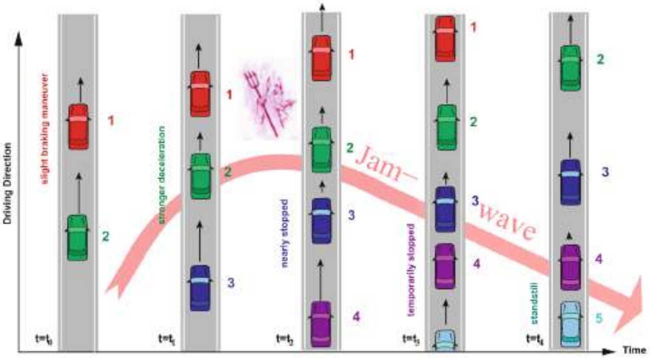
- See video.



Phantom Traffic Jam

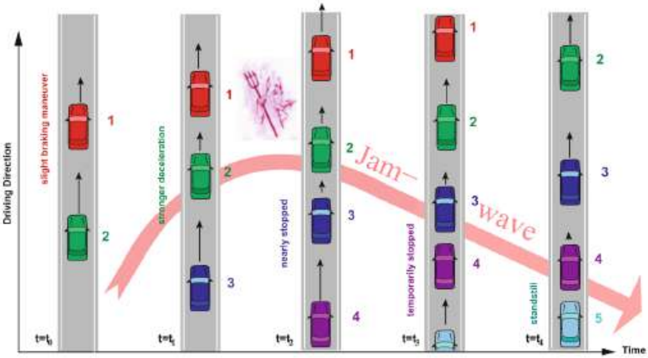
The phantom forms as follows (Note: one equilibrium state corresponds to a speed and a gap):

- All vehicles move with a equilibrium speed v_0 and gap $g(v_0)$;
- Vehicle 1 suddenly slightly decelerates from v_0 to v_1;



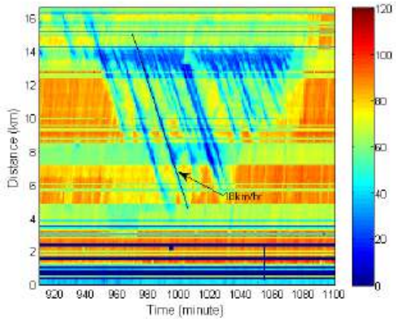
Phantom Traffic Jam

- When vehicle 2 realizes it (reaction time) and decelerates to v_1 from v_0, the gap has been smaller than $g(v_1)$
- To adapt the smaller gap, vehicle 2 has to decelerate to $v_2 < v_1$, and the gap becomes $g(v_2) < g(v_1)$
- Vehicle 3 has to decelerate to $v_3 < v_2$
- ....., until one vehicle has to stop

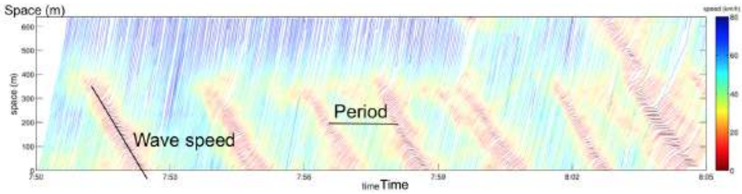


Oscillation or Stop-and-Go Wave

- From the perspective of a driver, it is:
  slowly move → decelerate → stop → accelerate → slowly move → ... (see video)
- From the macroscopic perspective, it is:



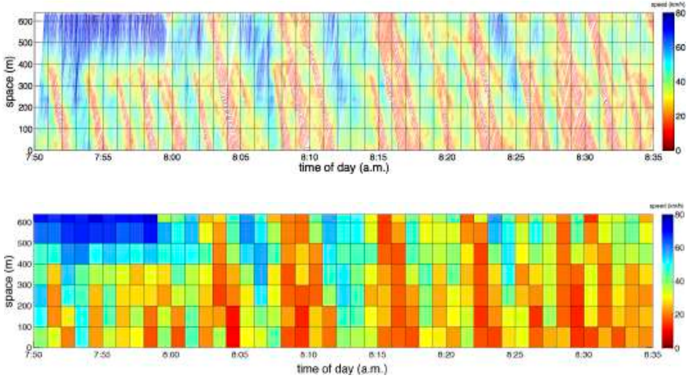
Oscillation or Stop-and-Go Wave



- **Main properties**:
  - Wave speed: -10 km/h ∼ -20 km/h
  - Period: 2 min ∼ 15 min
  - Amplitude (max speed-min speed): 20 ∼ 60 km/h
- **Side effects**: more fuels and emissions, make drivers tired and thus more danger.

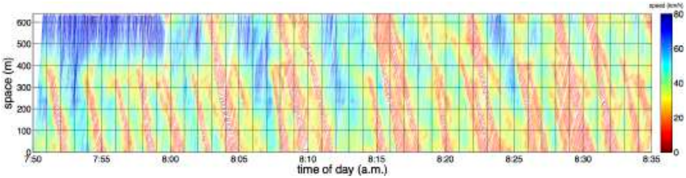
Time-Space Diagram

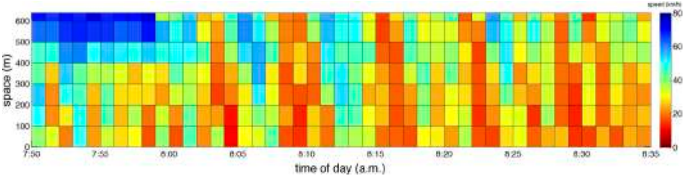
in which x-axis is time, y-axis is space, and the color inside (or z-axis) represents (aggregated) speed, is a foundational tool of various transportation research and applications.



Time-Space Diagram

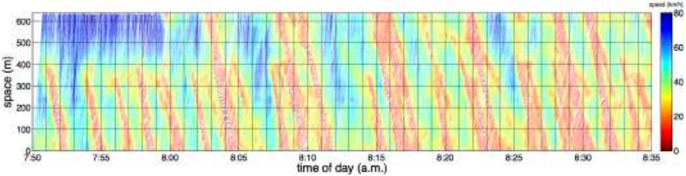
From a time-space diagram, we can identify traffic bottlenecks, understand traffic characteristics, predict travel time, and even estimate traffic emissions.

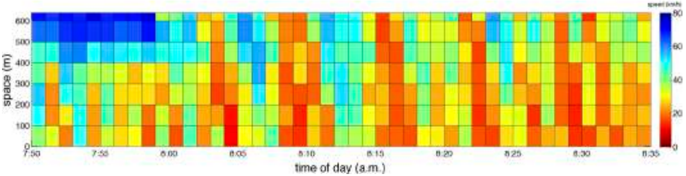




Time-Space Diagram

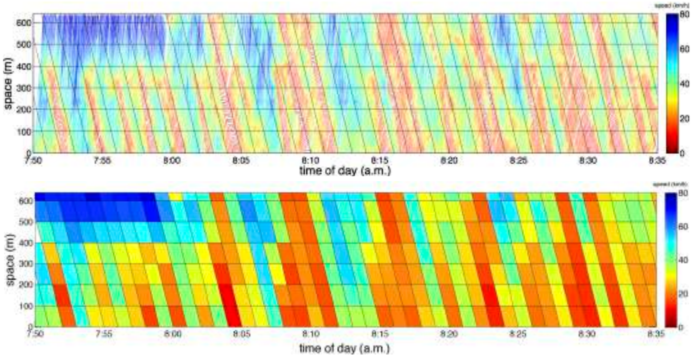
A time-space diagram could be constructed by using loop detector data, floating car (using navigation service) data, or high-fidelity vehicle trajectory data.





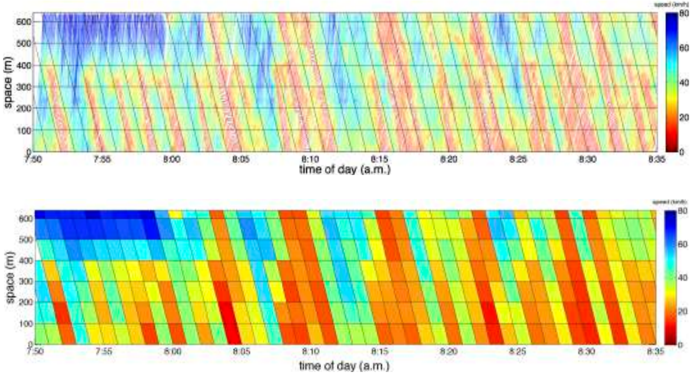
Time-Space Diagram: Research-1

Motivation: Traffic shows a clear pattern of propagation toward the lower-right direction, yet most studies still rely on the standard rectangular representation, which fails to capture this characteristic.



Zhengbing He et al. Constructing spatiotemporal speed contour diagrams: using rectangular or non-rectangular parallelogram cells, Transportmetrica B, 2019

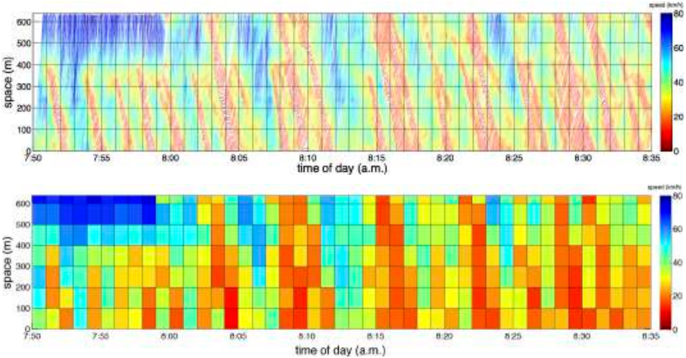
Contribution: This work empirically shows that *using non-rectangular parallelogram cells, which consider traffic waves*, to construct a time-space diagram is better.



Zhengbing He et al. Constructing spatiotemporal speed contour diagrams: using rectangular or non-rectangular parallelogram cells, Transportmetrica B, 2019

Time-Space Diagram: Research-2

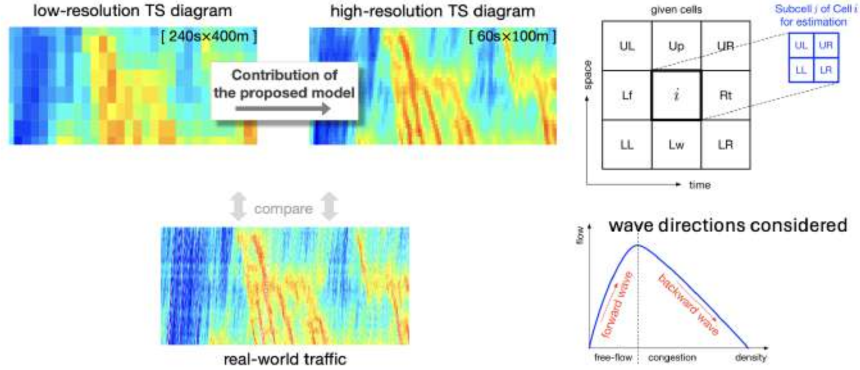
Motivation: The resolution of most time-space diagrams is quite coarse for detailed analyses and applications, mainly owing to the limitation of the existing information technology and traffic infrastructure investment.



Zhengbing He, Refining time-space traffic diagrams: A simple multiple linear regression model, IEEE Transactions on Intelligent Transportation Systems, 2024

Time-Space Diagram: Research-2

Contribution: This is the first work that introduces the refinement problem of the time-space traffic diagram and successfully solves it with only a small amount of high-fidelity traffic data.



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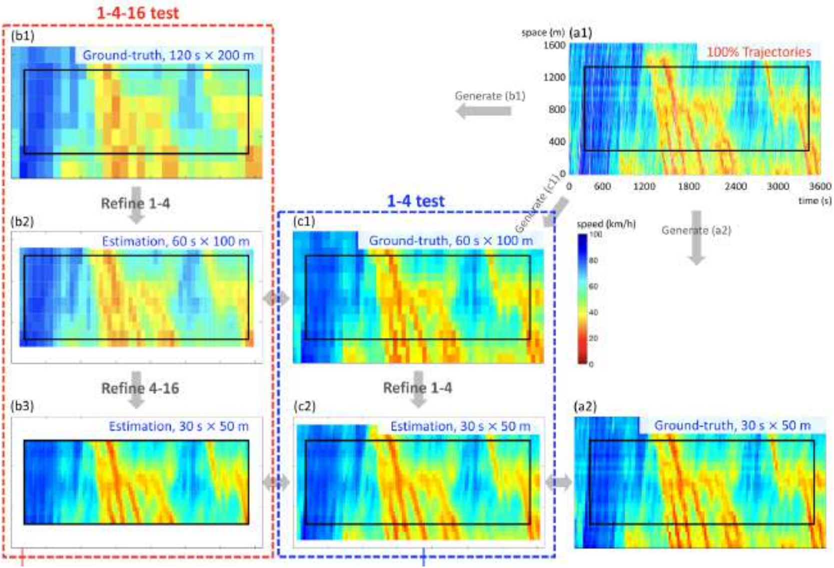
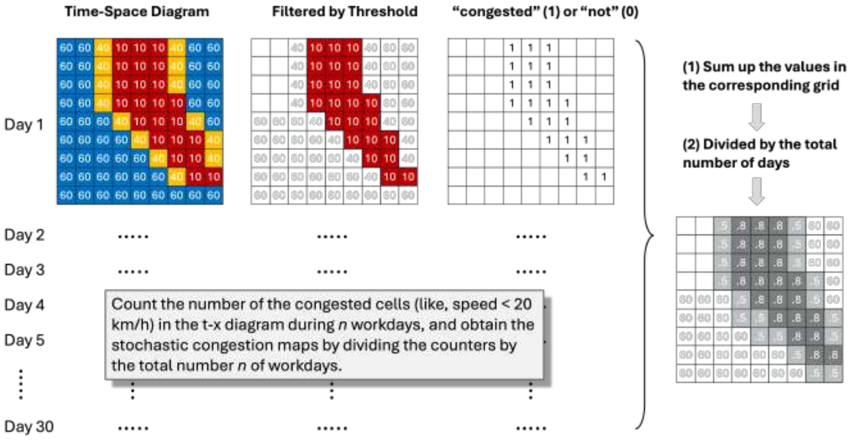
Time-Space Diagram: Research-2



Stochastic Congestion Map: Research-3

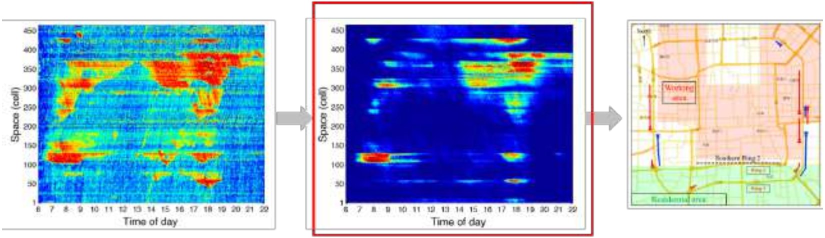
Represent the likelihood of the occurrence of congestion based on the observations for many days.



Stochastic Congestion Map: Research-3

Contribution

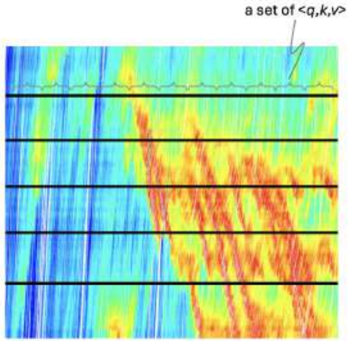
- Project to real map and visualize recurrent and non-recurrent congestion
- Only meaningful to predict traffic congestion whose occurrence probability is around 0.5
- A global value (e.g., average on all grids) is not appropriate



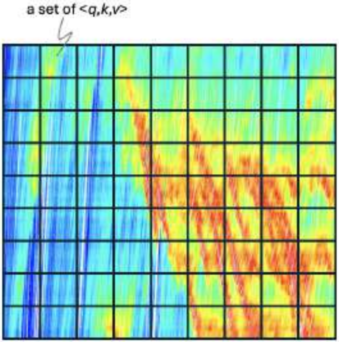
Zhengbing He et al, Mapping to Cells: A Simple Method to Extract Traffic Dynamics from Probe Vehicle Data, Computer-Aided Civil and Infrastructure Engineering, 2017

FD Construction: Research-4

Motivation: Given large-scale vehicle trajectory data, many researchers still extract macroscopic traffic variables using virtual loop detectors or rectangular time-space cells for data aggregation



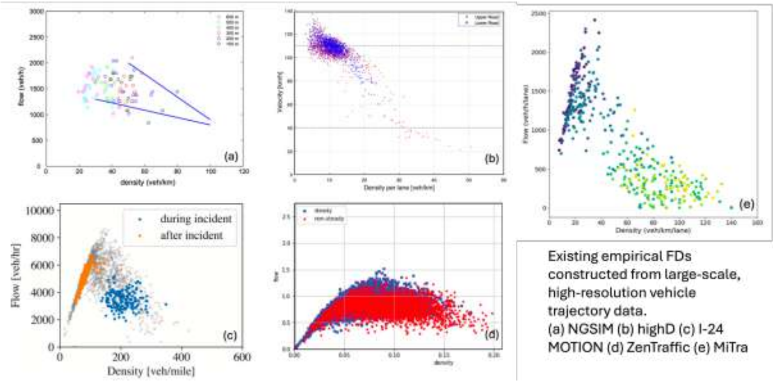
(virtual) loop detector



grid-based aggregation

Zhengbing He et al, Constructing the fundamental diagrams of traffic flow from large-scale vehicle trajectory data, arXiv, 2025

Motivation: Given large-scale vehicle trajectory data, many researchers still extract macroscopic traffic variables using virtual loop detectors or rectangular time-space cells for data aggregation

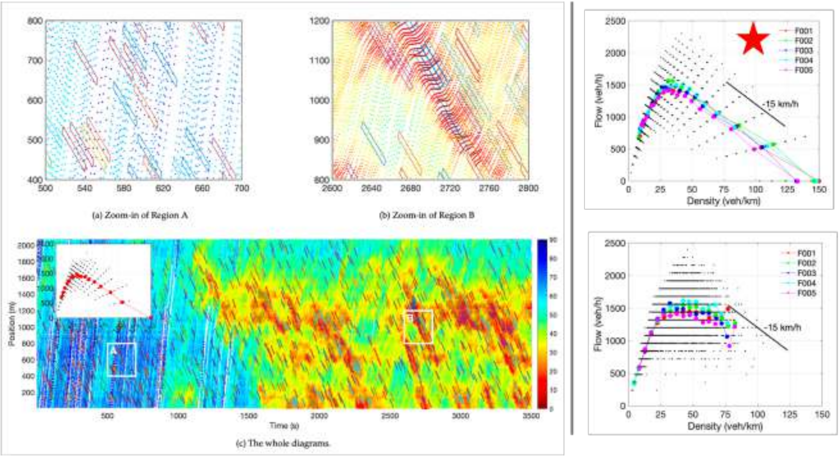


Existing empirical FDs constructed from large-scale, high-resolution vehicle trajectory data.
(a) NGSIM (b) highD (c) I-24 MOTION (d) ZenTraffic (e) MiTra

Zhengbing He et al, Constructing the fundamental diagrams of traffic flow from large-scale vehicle trajectory data, arXiv, 2025

FD Construction: Research-4

Contribution: A new tool that can automatically generate the fundamental diagram directly from a large-scale trajectory dataset



Zhengbing He et al, Constructing the fundamental diagrams of traffic flow from large-scale vehicle trajectory data, arXiv, 2025 (GitHub Link)

FD Construction: Research-4

Video

Thank you!